AC-DPO: Adaptive Capacity Direct Preference Optimization

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Abstract

Direct Preference Optimization (DPO) typically employs a fixed-capacity adapter throughout training, regardless of the varying difficulty of preference pairs. We propose Adaptive Capacity DPO (AC-DPO), a two-stage training framework that aligns model capacity with data difficulty by combining curriculum learning with dynamic LoRA rank expansion. In Stage 1, the model trains on easy preference pairs using a low-rank adapter ($r{=}8$, 295K parameters), then merges the learned weights into the base model. In Stage 2, a high-rank adapter ($r{=}64$, 2.36M parameters) is initialized for hard preference pairs. We evaluate AC-DPO against three baselines on a 10,000-pair subset of the Anthropic RLHF dataset using GPT-2. While all methods achieve comparable overall accuracy (43–44%), AC-DPO attains the highest token-level reward margin on hard tasks (+18% over the fixed $r{=}64$ baseline), confirming that capacity–difficulty alignment improves fine-grained discrimination on complex reasoning pairs. Ablation studies with a reverse curriculum and a fixed $r{=}8$ baseline further reveal that curriculum ordering interacts with capacity allocation in non-trivial ways, providing insights for future work on larger models.

1 Introduction

Aligning large language models (LLMs) with human preferences is a central challenge in modern NLP. Direct Preference Optimization [Rafailov et al., 2023] offers an efficient alternative to reinforcement learning from human feedback (RLHF) by directly optimizing a policy from preference pairs without training a separate reward model. In practice, DPO is often combined with parameter-efficient fine-tuning methods such as Low-Rank Adaptation (LoRA) [Hu et al., 2022], which inserts trainable low-rank matrices into frozen pretrained weights.

A key limitation of standard DPO with LoRA is that the adapter rank—and thus the model's trainable capacity—remains fixed throughout training. However, preference datasets are inherently heterogeneous: some pairs exhibit large quality gaps between chosen and rejected responses ("easy" pairs), while others require subtle reasoning to distinguish ("hard" pairs). Training a high-capacity adapter on easy data risks overfitting to superficial patterns, while a low-capacity adapter may lack the expressiveness needed for hard reasoning tasks.

Curriculum learning [Bengio et al., 2009] addresses data heterogeneity by ordering training examples from easy to hard. Recent work has applied curriculum strategies to DPO [Pattnaik et al., 2024], but these methods adjust only the *data* schedule while keeping model capacity fixed. Separately, dynamic rank methods have been explored in other domains [Deng et al., 2026, Croitoru et al., 2026], but not in the context of LLM preference optimization.

We propose **Adaptive Capacity DPO (AC-DPO)**, which jointly adapts both the data curriculum and the model's parameter capacity. Our contributions are:

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1. A two-stage training pipeline that combines curriculum-based data ordering with dynamic LoRA rank expansion, using adapter merging for knowledge transfer between stages.
2. An automated difficulty partitioning scheme using a pretrained reward model (DeBERTa-v3) to score and split preference pairs by reward margin.
3. A comprehensive evaluation including two ablation studies (reverse curriculum and fixed low-rank baseline) that isolate the effects of curriculum ordering and capacity expansion.

2 Related Works

Direct Preference Optimization. DPO [Rafailov et al., 2023] reformulates the RLHF objective by deriving a closed-form mapping between the reward function and the optimal policy, eliminating the need for a separate reward model during training. Given a preference dataset of triples $(\mathbf{x}, \mathbf{y}_w, \mathbf{y}_l)$ where $\mathbf{y}_w$ is preferred over $\mathbf{y}_l$, DPO directly optimizes the policy π_θ against a reference policy π_{ref}.

Low-Rank Adaptation. LoRA [Hu et al., 2022] parameterizes weight updates as $\Delta W = BA$ where $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$ with rank $r \ll \min(d, k)$. This reduces trainable parameters by orders of magnitude while preserving model quality. The choice of rank r controls the capacity–efficiency trade-off: higher ranks enable more expressive updates but increase computational cost and overfitting risk.

Curriculum Learning. Curriculum learning [Bengio et al., 2009] trains models on progressively harder examples. Curry-DPO [Pattnaik et al., 2024] applies this principle to preference optimization by ranking pairs by difficulty and training in easy-to-hard order, demonstrating improved alignment quality. However, Curry-DPO uses a fixed model capacity throughout training.

Dynamic Model Capacity. DR-LoRA [Deng et al., 2026] introduces dynamic rank expansion for mixture-of-experts in text-to-image generation, progressively increasing LoRA rank during training. Curriculum-DPO++ [Croitoru et al., 2026] combines data and model curricula for diffusion models. Our work extends this concept to LLM preference optimization, where the interaction between adapter capacity and preference pair difficulty has not been studied.

3 Background & Preliminaries

3.1 Direct Preference Optimization

Given a dataset $\mathcal{D} = \{(\mathbf{x}^{(i)}, \mathbf{y}_w^{(i)}, \mathbf{y}_l^{(i)})\}_{i=1}^N$ of prompts with preferred and dispreferred responses, DPO minimizes:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(\mathbf{x},\mathbf{y}_w,\mathbf{y}_l)\sim\mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(\mathbf{y}_w|\mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}_w|\mathbf{x})} - \beta \log \frac{\pi_\theta(\mathbf{y}_l|\mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}_l|\mathbf{x})} \right) \right] \tag{1}$$

where σ is the sigmoid function and β controls the deviation from the reference policy π_{ref}.

3.2 Low-Rank Adaptation

For a pretrained weight matrix $W_0 \in \mathbb{R}^{d\times k}$, LoRA introduces a low-rank update:

$$W = W_0 + \Delta W = W_0 + BA \tag{2}$$

where $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, and r is the adapter rank. The number of trainable parameters per adapted layer is $r(d + k)$, compared to dk for full fine-tuning.

3.3 Difficulty Scoring

We define the difficulty of a preference pair $(\mathbf{x}, \mathbf{y}_w, \mathbf{y}_l)$ using the reward margin from a pretrained reward model R:

$$m(\mathbf{x}, \mathbf{y}_w, \mathbf{y}_l) = R(\mathbf{x}, \mathbf{y}_w) - R(\mathbf{x}, \mathbf{y}_l) \tag{3}$$

Large positive margins indicate easy pairs where the quality difference is obvious; small or negative margins indicate hard pairs requiring nuanced judgment.

Algorithm 1 AC-DPO: Adaptive Capacity Direct Preference Optimization

Require: Base model θ_0, reference model π_{ref}, preference data $\mathcal{D}$, reward model R, ranks $r_1 < r_2$

Compute margins $m_i = R(\mathbf{x}_i, \mathbf{y}_{w,i}) - R(\mathbf{x}_i, \mathbf{y}_{l,i})$ for all i
Split $\mathcal{D}$ into $\mathcal{D}_{\text{easy}}$ (high margin) and $\mathcal{D}_{\text{hard}}$ (low margin) at median
Initialize LoRA adapter (B_1, A_1) with rank r_1 on θ_0
Train on $\mathcal{D}_{\text{easy}}$ with DPO loss $\rightarrow$ obtain trained (B_1, A_1)
Merge: $\theta_1 \leftarrow \theta_0 + B_1 A_1$
Initialize LoRA adapter (B_2, A_2) with rank r_2 on θ_1
Train on $\mathcal{D}_{\text{hard}}$ with DPO loss $\rightarrow$ obtain trained (B_2, A_2)
Merge: $\theta_{\text{final}} \leftarrow \theta_1 + B_2 A_2$
return θ_{final}

4 Method

AC-DPO consists of three phases: difficulty-based data partitioning, low-capacity training on easy data, and high-capacity training on hard data with knowledge transfer.

4.1 Data Partitioning

Given a preference dataset $\mathcal{D}$, we compute the reward margin m_i for each pair using a pretrained reward model (Equation 3). We sort pairs by margin in descending order and split at the median into:

$$\mathcal{D}_{\text{easy}} = \{(\mathbf{x}, \mathbf{y}_w, \mathbf{y}_l) \in \mathcal{D} \mid m \geq m_{\text{median}}\} \tag{4}$$

$$\mathcal{D}_{\text{hard}} = \{(\mathbf{x}, \mathbf{y}_w, \mathbf{y}_l) \in \mathcal{D} \mid m < m_{\text{median}}\} \tag{5}$$

4.2 Two-Stage Training with Capacity Expansion

Stage 1: Easy curriculum with low rank. We attach a LoRA adapter with rank r_1 (small) to the base model θ_0 and train on $\mathcal{D}_{\text{easy}}$ using the DPO objective (Equation 1). The low rank constrains the model to learn broad, high-signal patterns without overfitting.

Knowledge Transfer via Adapter Merging. After Stage 1, we permanently merge the learned low-rank weights into the base model:

$$\theta_1 = \theta_0 + B_1 A_1 \tag{6}$$

This preserves the knowledge from Stage 1 while freeing the adapter slot for Stage 2.

Stage 2: Hard curriculum with high rank. We initialize a new LoRA adapter with rank $r_2 \gg r_1$ on the merged model θ_1 and train on $\mathcal{D}_{\text{hard}}$. The higher rank provides the capacity needed for complex reasoning patterns, while the merged base already encodes basic alignment from Stage 1.

The complete procedure is summarized in Algorithm 1.

5 Experiments

5.1 Experimental Setup

Dataset. We use a 10,000-pair subset from the Anthropic RLHF dataset [Bai et al., 2022] (`Dahoas/rm-static`). Each pair consists of a prompt with a preferred (chosen) and dispreferred (rejected) response. We score all pairs using the DeBERTa-v3-large reward model [He et al., 2021] (`OpenAssistant/reward-model-deberta-v3-large-v2`) and split at the median reward margin into 5,000 easy pairs (margin ≥ 0.572) and 5,000 hard pairs (margin < 0.572). The easy subset has a mean margin of 1.698, while the hard subset has a mean margin of -0.151, indicating that many hard pairs have negative margins where the reward model disagrees with the human label. Each subset is further split 80/20 into train (4,000) and evaluation (1,000) sets with a fixed seed of 5100.

Base Model. We use GPT-2 (124M parameters) [Radford et al., 2019] as the base model, with a frozen copy serving as the reference model π_{ref}. All LoRA adapters target the `c_attn` (combined query-key-value) projection layers.

Table 1: Evaluation results on 1,000 easy, 1,000 hard, and 2,000 combined preference pairs. Best results per column are **bolded**. AC-DPO achieves the highest token margin on hard tasks, while all methods perform comparably on overall accuracy.

	Easy		Hard		All	
Method	Sum Acc	Tok Acc	Sum Acc	Tok Acc	Sum Acc	Tok Acc
AC-DPO (ours)	34.7	56.4	**51.8**	**52.6**	43.25	54.5
Baseline r=64	35.4	58.9	51.6	52.1	43.5	55.5
Baseline r=8	35.7	61.0	51.4	52.1	43.55	56.55
Reverse Curriculum	**36.1**	**62.8**	51.7	52.7	**43.9**	**57.75**

Table 2: Token-level reward margins (higher is better) and training efficiency. AC-DPO achieves the strongest discrimination on hard pairs while using fewer trainable parameters in Stage 1.

	Token Margin			Training Efficiency		
Method	Easy	Hard	All	Params	Runtime (s)	GPU (GB)
AC-DPO (ours)	0.146	**0.059**	0.103	295K→2.36M	2047	9.06
Baseline r=64	0.236	0.050	0.143	2.36M	2061	9.06
Baseline r=8	0.285	0.047	0.166	295K	2131	9.02
Reverse Curriculum	**0.331**	0.038	**0.185**	295K→2.36M	2122	9.06

Training Configuration. All experiments use $\beta = 0.1$, batch size 4, 4 training epochs, and the AdamW optimizer. AC-DPO Stage 1 uses learning rate 10^{-4} (rank r=8, 294,912 trainable parameters, 0.24% of total); Stage 2 uses learning rate 5×10^{-5} (rank r=64, 2,359,296 trainable parameters, 1.86% of total). Sequences are truncated to 512 tokens with a maximum prompt length of 256 tokens. All experiments use the same random seed for reproducibility.

Compared Methods. We evaluate four configurations to isolate the effects of curriculum ordering and capacity expansion:

- **AC-DPO** (ours): Easy→Hard curriculum with r=8 → r=64 expansion.
- **Baseline** r=64: Fixed r=64 on shuffled mixed data (standard DPO + LoRA).
- **Baseline** r=8: Fixed r=8 on shuffled mixed data (low-capacity control).
- **Reverse Curriculum**: Hard→Easy with r=8 → r=64 (ablation for curriculum direction).

Evaluation Metrics. For each evaluation pair, we compute the log-probability of the chosen and rejected responses under the trained model. We report:

- **Sum Accuracy**: fraction of pairs where $\sum \log p(\mathbf{y}_w|\mathbf{x}) > \sum \log p(\mathbf{y}_l|\mathbf{x})$.
- **Token Accuracy**: fraction where the average per-token log-probability favors the chosen response.
- **Token Margin**: mean difference in per-token log-probabilities, $\bar{\log} p(\mathbf{y}_w) - \bar{\log} p(\mathbf{y}_l)$, measuring the model's confidence in its preference.

5.2 Main Results

Table 1 presents the evaluation results across all four methods.

Overall accuracy is comparable across methods. All four methods achieve overall sum accuracy in the narrow range of 43.25–43.9%, suggesting that at the GPT-2 scale with 10K training pairs, the choice of curriculum ordering and adapter rank has limited impact on aggregate preference accuracy. This is consistent with the observation that GPT-2's base capacity may be the primary bottleneck.

AC-DPO achieves the strongest hard-task discrimination. On hard preference pairs, AC-DPO attains the highest token margin (0.059), exceeding the fixed r=64 baseline by 18% (0.050) and the

Training loss curves: AC-DPO (Stage 1 + Stage 2), Baseline r=64, Reverse Curriculum (Stage 1 + Stage 2). AC-DPO Stage 1 converges fastest on easy data; Stage 2 starts high due to new adapter initialization on hard data.

Figure 1: Training loss over steps. AC-DPO Stage 1 (easy, r=8) converges rapidly to 0.464. Stage 2 (hard, r=64) begins at $\sim$1.1 after adapter re-initialization and decreases to 0.715. The baseline on mixed data converges to 0.624.

fixed r=8 baseline by 26% (0.047). This indicates that dedicating high capacity specifically to hard data—after first learning basic patterns with low capacity—produces sharper per-token distinctions on complex reasoning pairs.

Easy-task performance trades off with hard-task focus. AC-DPO shows lower easy-task token accuracy (56.4%) compared to baselines (58.9–62.8%). This is expected: Stage 1 uses only r=8 (295K parameters) on easy data, providing less capacity than the r=64 baselines. The merged Stage 1 knowledge may also be partially overwritten during Stage 2 training on hard data.

5.3 Ablation Studies

Effect of curriculum direction (Reverse Curriculum). The reverse curriculum (Hard→Easy) achieves the highest overall accuracy (43.9%) and token accuracy (57.75%), outperforming AC-DPO's easy-to-hard ordering. This initially surprising result can be explained by *recency bias*: the model's final performance is disproportionately influenced by the last training stage. The reverse curriculum dedicates its high-capacity Stage 2 to easy data, which has clearer preference signals and constitutes half the evaluation set. However, its hard-task token margin (0.038) is the lowest among all methods, confirming that it sacrifices hard-task discrimination for easy-task performance—the opposite trade-off from AC-DPO.

Effect of adapter capacity (Baseline r=8). The fixed r=8 baseline performs comparably to the r=64 baseline on sum accuracy (43.55% vs 43.5%) and even outperforms it on token accuracy (56.55% vs 55.5%). This suggests that at the GPT-2 scale, the low-rank adapter may provide beneficial regularization. However, the r=8 baseline has the lowest hard-task token margin (0.047), indicating that while low capacity suffices for coarse preference accuracy, it limits fine-grained discrimination on difficult pairs.

5.4 Training Dynamics

Figure 1 illustrates the training loss trajectories. AC-DPO Stage 1 converges rapidly on easy data (final loss 0.464), reflecting the clear preference signals. Stage 2 starts with elevated loss ($\sim$1.1) due to the newly initialized high-rank adapter and harder data, then decreases to 0.715. The baseline trains on mixed data with a smoother trajectory (final loss 0.624). The reverse curriculum shows the opposite pattern: Stage 1 on hard data converges slowly (loss 0.572), while Stage 2 on easy data converges faster (loss 0.638).

5.5 Compute Efficiency Analysis

Table 3 compares the effective parameter-compute across methods. We define parameter-steps as the product of trainable parameters and training steps, measuring the total optimization work.

AC-DPO uses only 56% of the parameter-steps of the r=64 baseline while achieving comparable accuracy and superior hard-task discrimination. The wall-clock times are similar because GPU utilization is dominated by forward/backward passes through the full model rather than the adapter parameters alone. On larger models where adapter computation becomes a larger fraction of total cost, the parameter-step savings would translate more directly to wall-clock savings.

Table 3: Compute efficiency comparison. AC-DPO uses 55% of the parameter-steps of the fixed r=64 baseline by leveraging low-rank training in Stage 1.

Method	**Param-Steps** ($\times 10^9$)	**Relative**	**Wall Time (s)**
AC-DPO (ours)	$0.295\text{M} \times 4000 + 2.36\text{M} \times 4000 = 10.6$	0.56×	2047
Baseline r=64	$2.36\text{M} \times 8000 = 18.9$	1.00×	2061
Baseline r=8	$0.295\text{M} \times 8000 = 2.4$	0.13×	2131
Reverse Curriculum	$0.295\text{M} \times 4000 + 2.36\text{M} \times 4000 = 10.6$	0.56×	2122

6 Conclusions

We presented AC-DPO, a framework that aligns model capacity with data difficulty during preference optimization by combining curriculum learning with dynamic LoRA rank expansion. Our two-stage pipeline—low-rank training on easy pairs followed by high-rank training on hard pairs with adapter merging for knowledge transfer—achieves the highest token-level reward margin on hard preference pairs (+18% over the fixed-rank baseline) while using only 56% of the parameter-compute budget. Although overall accuracy differences are small at the GPT-2 scale, the consistent advantage in hard-task discrimination validates the core hypothesis that matching capacity to difficulty improves fine-grained preference learning.

Our ablation studies reveal two additional insights: (1) curriculum direction matters—the reverse curriculum trades hard-task discrimination for easy-task performance, confirming that the easy-to-hard ordering in AC-DPO specifically benefits complex reasoning; (2) at the GPT-2 scale, low-rank adapters provide implicit regularization that can match high-rank adapters on coarse accuracy, but not on fine-grained token-level margins.

7 Limitations & Future Work

Model scale. Our experiments use GPT-2 (124M parameters), where the base model's capacity may be the primary bottleneck, compressing all methods into a narrow accuracy range. The benefits of dynamic capacity allocation are expected to be more pronounced on larger models (e.g., LLaMA-7B or 13B) where the gap between r=8 and r=64 represents a more meaningful fraction of the model's expressiveness.

Binary difficulty split. We partition data into only two difficulty levels (easy/hard) at the median. A finer-grained, multi-stage curriculum with gradual rank expansion (e.g., r=8 $\rightarrow$ 16 $\rightarrow$ 32 $\rightarrow$ 64) could provide smoother capacity–difficulty alignment and reduce the loss spike observed at stage transitions.

Catastrophic forgetting. AC-DPO's lower easy-task performance suggests that Stage 2 training partially overwrites Stage 1 knowledge despite adapter merging. Techniques such as experience replay (mixing easy examples into Stage 2) or elastic weight consolidation could mitigate this forgetting.

Evaluation scope. We evaluate only log-probability-based preference accuracy. Future work should include generation-quality metrics (e.g., win rates from GPT-4 judges, human evaluation) and downstream task performance to assess whether improved token margins translate to better generated outputs.

Recency bias. The reverse curriculum's strong overall performance highlights a recency bias in two-stage training: the final stage disproportionately influences evaluation metrics. Investigating training schedules that interleave difficulty levels or use replay buffers could address this limitation.

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A Detailed Per-Step Training Metrics

Table 4 reports training metrics at key checkpoints for all methods.

Table 4: Training loss and reward accuracy at selected steps.

Method	**Checkpoint**	**Loss**	**Reward Acc**	**Margin**
AC-DPO Stage 1	Step 1000	0.580	0.775	0.733
(r=8, easy)	Step 2000	0.420	0.750	1.620
	Step 4000	0.597	0.725	1.246
AC-DPO Stage 2	Step 1000	0.748	0.525	−0.038
(r=64, hard)	Step 2000	0.686	0.475	0.053
	Step 4000	0.663	0.650	0.112
Baseline r=64	Step 2000	0.617	0.625	0.268
(mixed)	Step 4000	0.684	0.600	0.116
	Step 8000	0.680	0.475	0.171
Reverse Stage 1	Step 2000	0.669	0.575	0.070
(r=8, hard)	Step 4000	0.572	0.775	0.341
Reverse Stage 2	Step 2000	0.475	0.750	0.959
(r=64, easy)	Step 4000	0.639	0.625	0.661
Baseline r=8	Step 2000	0.618	0.650	0.322
(mixed)	Step 4000	0.738	0.600	0.116
	Step 8000	0.769	0.500	0.145

B Experimental Infrastructure

All experiments were conducted on a single NVIDIA A40 GPU (48GB VRAM) on the WashU Academic Compute Cluster. Peak GPU memory usage was approximately 9.06 GB across all configurations. The complete experimental pipeline (data preparation, four training runs, and evaluation) required approximately 2.5 hours of GPU time. Software stack: Python 3.10, PyTorch, Hugging Face Transformers, PEFT, and TRL.

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